

# Course Content Simplification Agent

## Problem Statement

### The Challenge

Educational materials such as textbooks, lecture notes, research papers, and faculty resources often contain complex terminology and advanced explanations that are difficult for many students to understand. Learners with different levels of prior knowledge struggle to grasp important concepts due to jargon-heavy language and lack of personalized explanations. Existing educational platforms provide the same content to every learner without adapting it to their individual learning needs, resulting in reduced comprehension and lower learning efficiency.

## The Objective

Develop an AI-powered Course Content Simplification Agent that delivers personalized learning support by adapting educational content according to the learner's proficiency level. The solution should:

- Simplify complex academic content into easy-to-understand language.
- Generate concise summaries highlighting the key concepts.
- Explain difficult words and technical terms in simple language.
- Create flashcards for quick revision and knowledge retention.
- Generate multiple-choice quizzes for self-assessment.
- Support different learner levels such as Beginner, Intermediate, High School, Undergraduate, Engineering Student, and Expert.